

Module 20: Power BI Dataflows & Reusable Data Preparation



Overview

In many organizations, multiple reports use the same data transformation logic. If each report performs its own data cleaning and transformation, it creates duplication, inconsistency, and maintenance challenges.

Power BI Dataflows solve this problem by centralizing data preparation in the Power BI Service.

Dataflows enable organizations to create reusable ETL (Extract, Transform, Load) processes, standardize business logic, improve data quality, and reduce development effort across multiple reports and datasets.

This module covers Dataflows, Power Query Online, Linked Tables, Computed Tables, Dataflow Governance, Performance Optimization, and Enterprise Data Preparation strategies.



Learning Objectives

After completing this module, you will be able to:

- Understand Power BI Dataflows
 - Understand Dataflow architecture
 - Create and manage Dataflows
 - Use Power Query Online
 - Build reusable ETL processes
 - Create Linked Tables
 - Create Computed Tables
 - Improve data consistency
 - Apply Dataflow governance
 - Design enterprise data preparation solutions
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Table of Contents

1. Introduction to Dataflows
2. Why Dataflows Matter
3. Traditional ETL vs Dataflows
4. Dataflow Architecture
5. Power Query Online
6. Creating a Dataflow
7. Dataflow Storage
8. Entities in Dataflows
9. Linked Tables
10. Computed Tables

11. Reusable ETL Design
12. Dataflow Refresh
13. Incremental Refresh in Dataflows
14. Dataflow Governance
15. Dataflow Performance Optimization
16. Enterprise Data Architecture
17. Best Practices
18. Common Mistakes
19. Key Terminologies
20. Practice Exercises
21. Mini Project
22. Module Summary

1 Introduction to Dataflows

What is a Dataflow?

A Dataflow is a cloud-based data preparation and transformation layer in Power BI Service.

Purpose

Create reusable data transformation processes.

Dataflow Workflow

```
graph TD;
    A[Data Sources] --> B[Dataflow];
    B --> C[Semantic Model];
    C --> D[Report];
```

Benefits

- Centralized ETL
 - Reusable Transformations
 - Consistent Business Logic
-

2 Why Dataflows Matter

Without Dataflows:

```
Report A
  ↓
Own ETL

Report B
  ↓
Own ETL

Report C
  ↓
Own ETL
```

Problems

Duplicate Logic

Increased Maintenance

Inconsistent Calculations

Slower Development

With Dataflows

```
Dataflow
  ↓
Shared Clean Data
  ↓
Multiple Reports
```

3 Traditional ETL vs Dataflows

Traditional ETL

Data preparation occurs inside each report.

Dataflow Approach

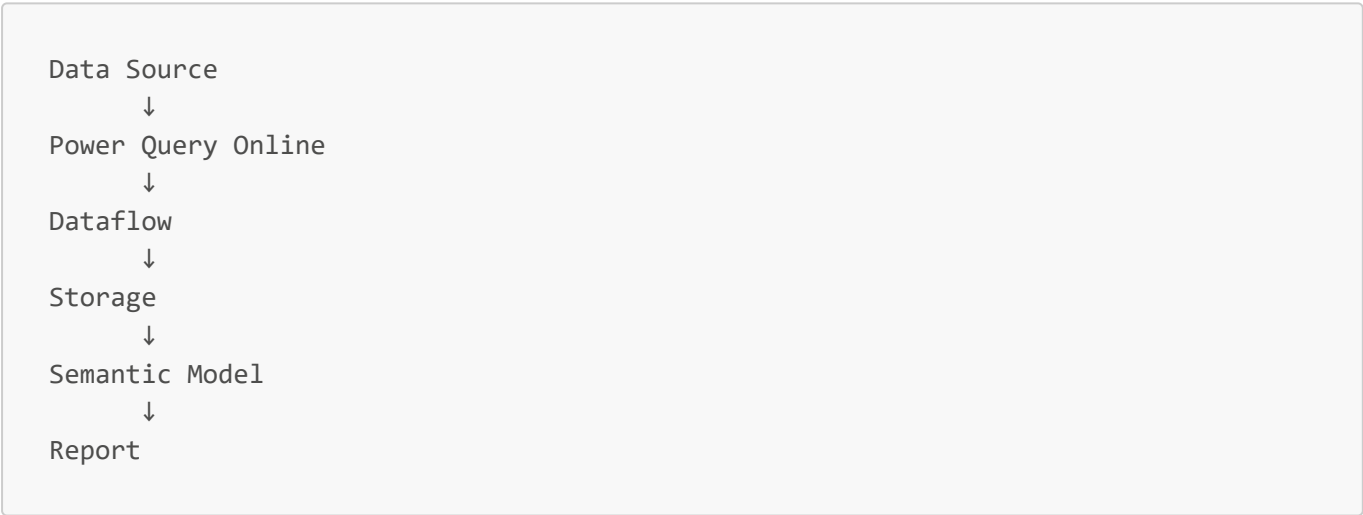
Data preparation occurs once.

Comparison

Feature	Traditional ETL	Dataflow
Reusability	Low	High
Maintenance	Difficult	Easier
Consistency	Lower	Higher
Scalability	Limited	Better
Governance	Difficult	Easier

 Dataflow Architecture

High-Level Architecture



Components

Source Systems

SQL Server

Excel

CSV

APIs

ERP Systems

Transformation Layer

Power Query Online

Storage Layer

Dataflow Storage

Consumption Layer

Datasets and Reports

Power Query Online

What is Power Query Online?

Cloud-based version of Power Query.

Similarities to Desktop

Supports:

- M Language
 - Transformations
 - Data Cleaning
 - Data Shaping
-

Advantages

Browser-Based

No Desktop required.

Centralized

Shared transformations.

Reusable

Used across reports.

Creating a Dataflow

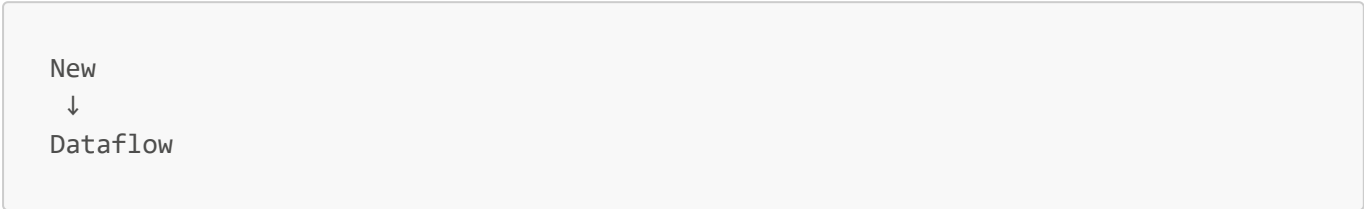
Steps

Step 1

Open Workspace.

Step 2

Select:



New
↓
Dataflow

Step 3

Choose Data Source.

Step 4

Apply Transformations.

Step 5

Save Dataflow.

Step 6

Configure Refresh.

Dataflow Storage

Where is Data Stored?

Dataflows store processed data in the Power BI cloud environment.

Benefits

Centralized Storage

Reusability

Improved Governance

Architecture

```
Raw Data
  ↓
Transformation
  ↓
Cloud Storage
  ↓
Reports
```

8 Entities in Dataflows

What is an Entity?

An Entity is similar to a table.

Example

Customer Entity

CustomerID	CustomerName
------------	--------------

Product Entity

ProductID	ProductName
-----------	-------------

Purpose

Store transformed data.

9 Linked Tables

What are Linked Tables?

Linked Tables reference entities from another Dataflow.

Example

```
Master Dataflow
  ↓
Customer Entity
  ↓
Linked Dataflow
```

Benefits

Avoid Duplication

Centralized Maintenance

Better Governance

10 Computed Tables

What are Computed Tables?

Tables created from existing Dataflow entities.

Example

```
Sales Entity
  ↓
Aggregation
  ↓
Monthly Sales Table
```

Benefits

Reusable Business Logic

Faster Development

Improved Performance

1 1 Reusable ETL Design

Goal

Transform data once and reuse everywhere.

Example

Instead of:

```
Sales Report
  ↓
Clean Customer Names

Finance Report
  ↓
Clean Customer Names

HR Report
  ↓
Clean Customer Names
```

Use:

```
Dataflow
  ↓
Clean Customer Names
  ↓
All Reports
```

Benefits

Consistency

Reduced Maintenance

Faster Development

1 2 Dataflow Refresh

Purpose

Keep data current.

Refresh Types

Manual Refresh

Scheduled Refresh

On-Demand Refresh

Workflow

```
graph TD; A[Source Data] --> B[Refresh]; B --> C[Updated Dataflow]; C --> D[Updated Reports];
```

1 3 Incremental Refresh in Dataflows

What is Incremental Refresh?

Refresh only changed data.

Example

Historical Data

```
2021
2022
2023
2024
```

Refresh only:

```
2024
```

Benefits

Faster Refreshes

Lower Resource Usage

Better Scalability

Dataflow Governance

Why Governance Matters

Dataflows often become enterprise assets.

Governance Areas

Ownership

Assign responsible teams.

Documentation

Describe transformations.

Security

Control access.

Monitoring

Track usage.

Goal

Create trusted data assets.

Dataflow Performance Optimization

Best Practices

Remove Unnecessary Columns

Filter Early

Reduce data volume.

Use Incremental Refresh

Avoid Redundant Transformations

Optimize Source Queries

Benefits

Faster Refresh

Better Scalability

Reduced Costs

1 **6** Enterprise Data Architecture

Recommended Architecture

```
graph TD;
    A[Source Systems] --> B[Dataflows];
    B --> C[Semantic Models];
    C --> D[Reports];
    D --> E[Dashboards];
```

Benefits

Standardization

Reusability

Scalability

Governance

1 **7** Best Practices

Create Shared Dataflows

Use Consistent Naming

Document Transformations

Establish Ownership

Monitor Refreshes

Reuse Existing Dataflows

Apply Governance Policies

1 **8** Common Mistakes

Duplicate Dataflows

Poor Documentation

No Ownership

Excessive Transformations

Ignoring Refresh Failures

Rebuilding Existing Logic



Key Terminologies

Term	Definition
Dataflow	Reusable cloud-based ETL process
ETL	Extract, Transform, Load
Entity	Table within a Dataflow
Linked Table	Reference to another Dataflow entity
Computed Table	Derived Dataflow table
Power Query Online	Cloud-based transformation engine
Refresh	Updating Dataflow data
Incremental Refresh	Refreshing only changed data
Governance	Managing Dataflow usage
Reusable ETL	Shared transformation process



Practice Exercises

Exercise 1

Create a Dataflow using a CSV source.

Exercise 2

Create Customer and Product entities.

Exercise 3

Configure scheduled refresh.

Exercise 4

Design a reusable ETL architecture.

Exercise 5

Compare:

- Dataflows
- Power Query in Desktop



Mini Project

Enterprise Sales Dataflow

Scenario

Multiple departments use sales data.

Requirements

Data Sources

- Sales Data
 - Customer Data
 - Product Data
-

Transformations

- Clean Customer Names
 - Standardize Product Categories
 - Remove Duplicates
-

Dataflow

Create reusable entities.

Governance

Define:

- Ownership
 - Documentation
 - Refresh Schedule
-

Deliverables

1. Dataflow Design
 2. Entity Structure
 3. Refresh Strategy
 4. Governance Plan
-



Module Summary

In this module, you learned:

- ✓ Dataflows
- ✓ Dataflow Architecture
- ✓ Power Query Online
- ✓ Dataflow Storage
- ✓ Entities
- ✓ Linked Tables
- ✓ Computed Tables
- ✓ Reusable ETL
- ✓ Dataflow Refresh
- ✓ Incremental Refresh
- ✓ Dataflow Governance
- ✓ Performance Optimization
- ✓ Enterprise Data Architecture

Dataflows provide a scalable, reusable, and governed approach to data preparation. They are a key component of enterprise Power BI architectures and help organizations create a single source of trusted, reusable data.



Next Module

Module 21: Power BI Data Modeling Masterclass

Topics Covered:

- Advanced Data Modeling
- Fact & Dimension Design
- Surrogate Keys
- Degenerate Dimensions
- Slowly Changing Dimensions (SCD)
- Bridge Tables
- Many-to-Many Modeling
- Role-Playing Dimensions
- Composite Models
- Enterprise Data Warehouse Design